AI5707 – Pattern Recognition and Machine Learning

Assignment 6: Generation of Data Using 2D Gaussian Mixture Model, K-Means Clustering and Classification

October 16, 2026

Name : KARTHIKEYAN P L
Roll No. : 26120031

1 Introduction

This assignment studies two-dimensional data generation using Gaussian mixture models (GMMs), K-Means clustering, and distortion-based classification. Two classes are generated, each represented by a mixture of three Gaussian components. Spherical, diagonal, and full covariance matrices are compared. K-Means is fitted separately to the training observations of each class, and held-out observations are classified using minimum squared distortion to the class-specific centroids.

The mixture density is

$$p(\mathbf{x}) = \sum_{i=1}^{K} \pi_i \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_i, C_i), \qquad \sum_{i=1}^{K} \pi_i = 1,$$

where $\mathbf{x} = [x_1, x_2]^T$, π_i is the mixture weight, $\boldsymbol{\mu}_i$ is the component mean, and C_i is its covariance matrix.

2 Gaussian Mixture Data Generation

Each class contains 1,500 observations generated from a three-component Gaussian mixture. The component means are vertically arranged at $x_2 = 3.5, 0, -3.5$. Class 1 is centred near $x_1 = 0$, and Class 2 near $x_1 = 7$. The mixture weights are $[0.40, 0.20, 0.40]$.

2.1 Covariance configurations

Spherical covariance $C_i = \sigma_i^2 I$ produces approximately circular clouds. Diagonal covariance allows different variances along the coordinate axes. Full covariance includes non-zero off-diagonal entries and can produce tilted elliptical clouds.

Table 1: Covariance matrices used in the experiment

Type	Components 1 and 3	Component 2
Spherical	$0.65^2 I$	$0.55^2 I$
Diagonal	$\begin{bmatrix} 1.00 & 0 \\ 0 & 0.40 \end{bmatrix}$	$\begin{bmatrix} 0.45 & 0 \\ 0 & 0.80 \end{bmatrix}$
Full	$\begin{bmatrix} 0.80 & 0.30 \\ 0.30 & 0.50 \end{bmatrix}$	$\begin{bmatrix} 0.65 & -0.20 \\ -0.20 & 0.70 \end{bmatrix}$

2.2 Gaussian sampling

The Box–Muller transform converts independent uniform random variables $U_1, U_2 \in (0, 1)$ into standard normal samples:

$$Z_0 = \sqrt{-2 \ln U_1} \cos(2\pi U_2), \quad Z_1 = \sqrt{-2 \ln U_1} \sin(2\pi U_2).$$

The generator uses the eigendecomposition $C = V \Lambda V^T$ to transform standard-normal samples. It checks that each covariance matrix is 2×2, symmetric, and positive definite. Cholesky decomposition is not used.

3 Part A: Synthetic Gaussian Data

3.1 Generated distributions

Figure 1 shows the two classes under spherical, diagonal, and full covariance assumptions. Colours distinguish the three Gaussian mixture components.

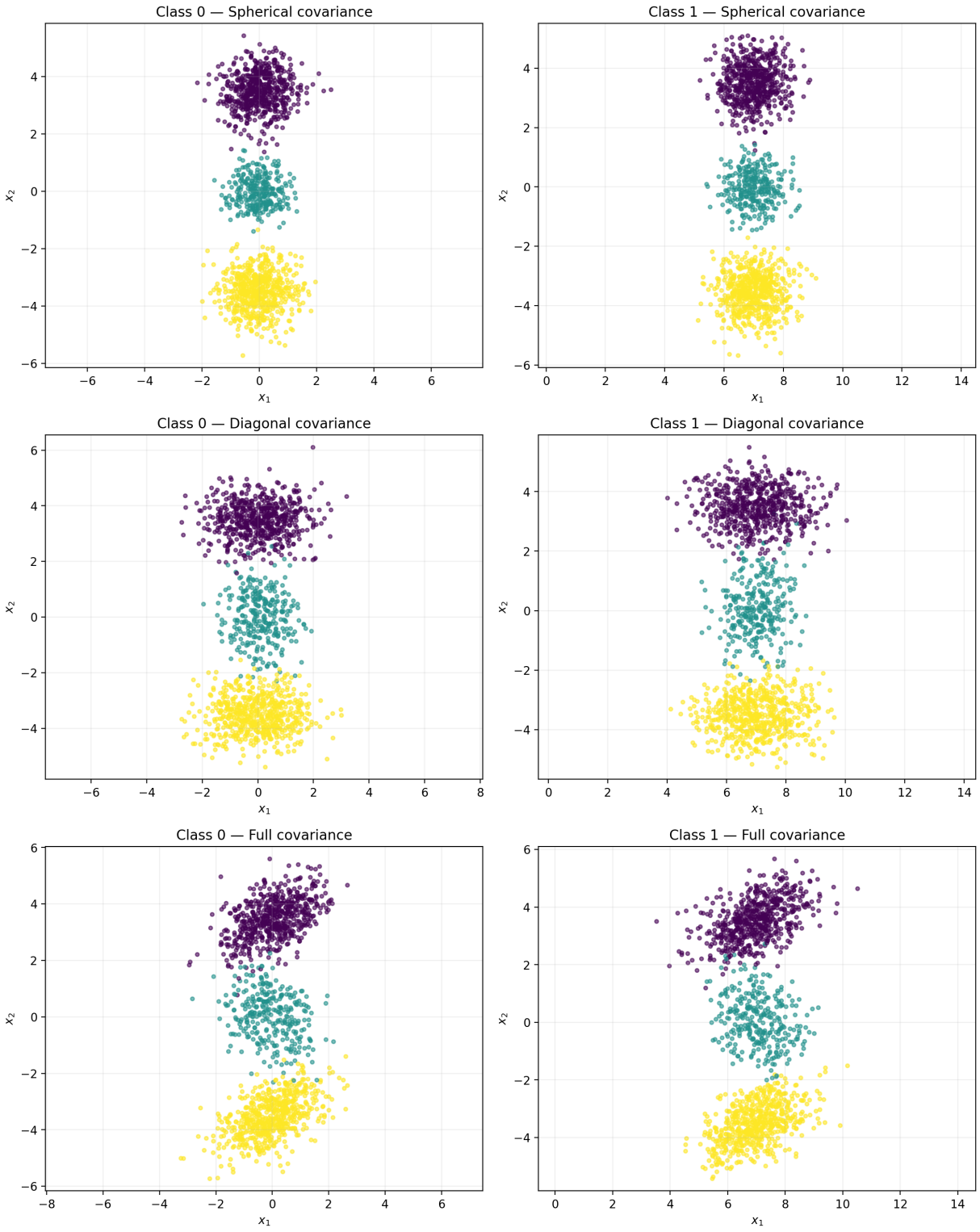


Figure 1: Generated samples using spherical, diagonal, and full covariance matrices.

3.2 Train–test split

The full-covariance data are used for the main classification experiment. Each class is split independently in an 80:20 ratio: 1,200 training and 300 test observations per class. The combined training set contains 2,400 observations and the combined test set contains 600 observations.

3.3 K-Means clustering

K-Means is implemented from scratch. It assigns each training point to the nearest centroid using squared Euclidean distance, updates each centroid to the mean of assigned points, and repeats until the centroid shift is below the tolerance or the iteration limit is reached. The notebook uses $K = 2$ centroids per class, distinct from the three Gaussian components used for generation.

Figure 2 shows within-class nearest-centroid regions. These are Voronoi regions within each class, not the final boundary separating the two classes. Red crosses indicate learned centroids.

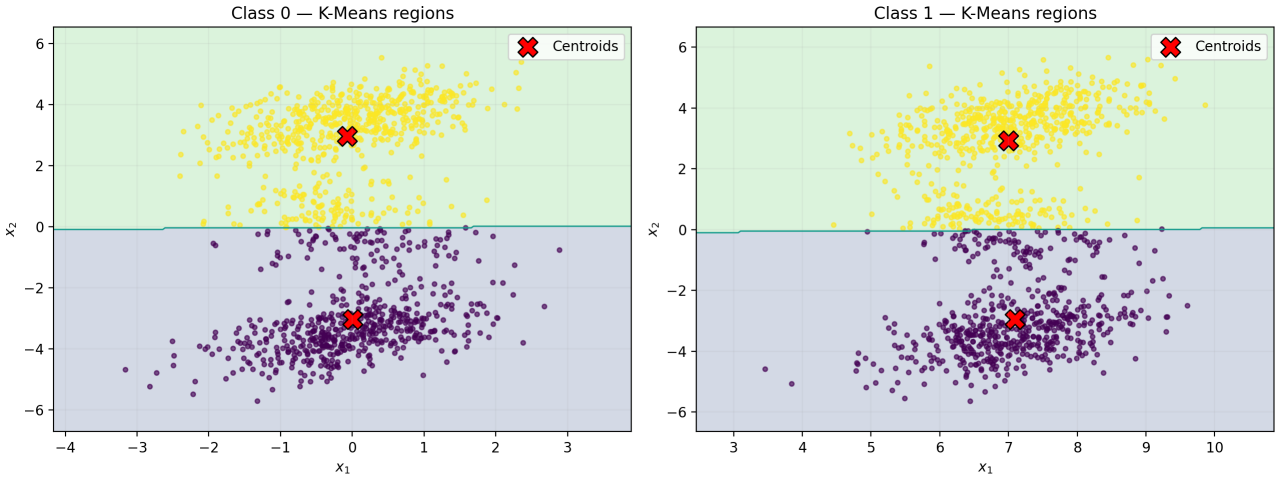


Figure 2: K-Means regions and centroids fitted separately to each class's training data.

3.4 Distortion-based classification

For a test point $\mathbf{x}$, distortion to class c is the minimum squared distance to its centroids:

$$D_c(\mathbf{x}) = \min_k \|\mathbf{x} - \boldsymbol{\mu}_{c,k}\|^2.$$

The predicted class is the one with the smaller distortion, $\hat{y} = \arg\min_c D_c(\mathbf{x})$. The final decision boundary consists of points satisfying $D_0(\mathbf{x}) = D_1(\mathbf{x})$.

4 Results

4.1 Main test-set performance

The recorded full-covariance run correctly classified all 600 held-out observations, giving 100.00% accuracy.

Table 2: Confusion matrix for the full-covariance experiment

True class	Predicted Class 1	Predicted Class 2
Class 1	300	0
Class 2	0	300

$$\text{Accuracy} = \frac{600}{600} = 1.0000 = 100.00\%.$$

4.2 Test predictions

Figure 3 reproduces the instructor-provided reference image. Blue points represent Class 1, orange points represent Class 2, and the black line marks equal minimum distortion to the two class-specific centroid sets.

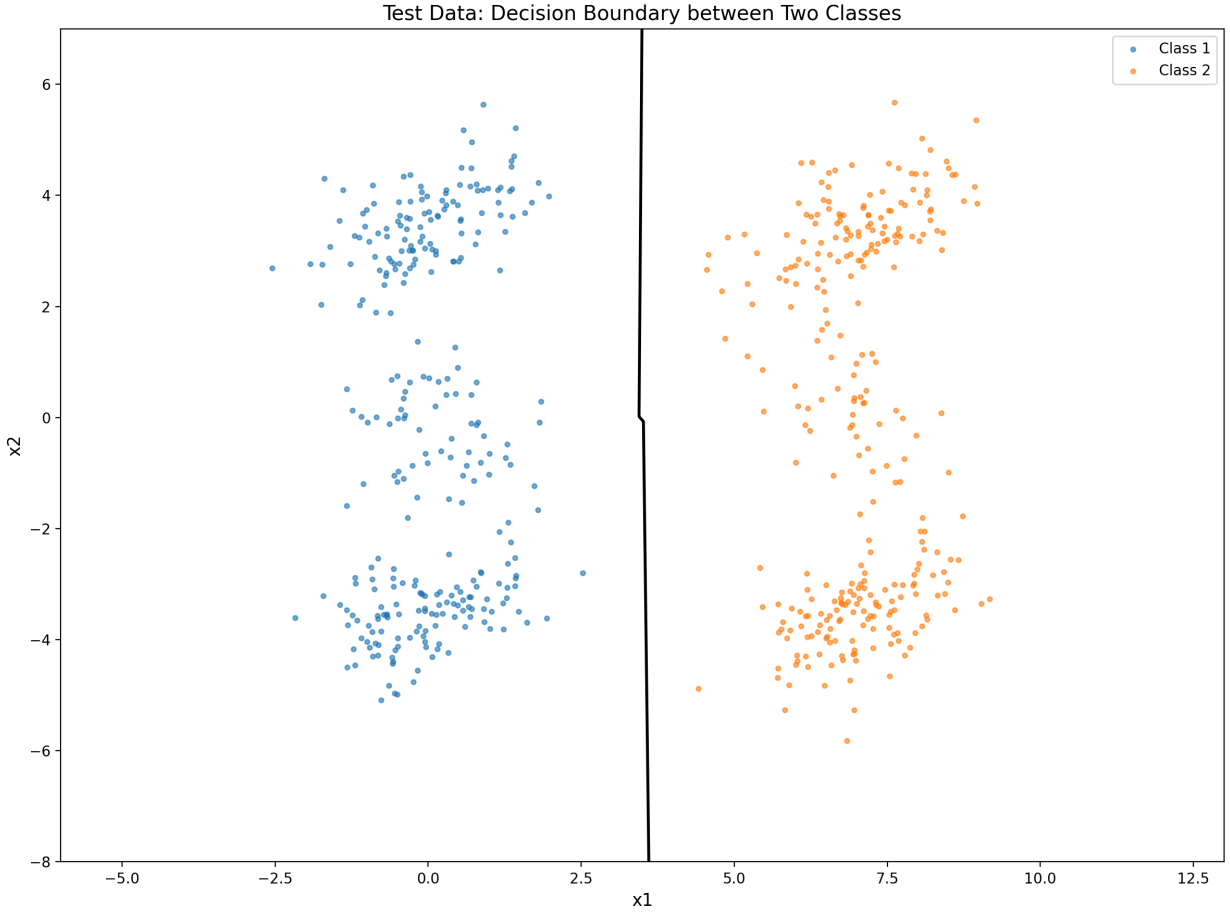


Figure 3: Notebook-generated test data and decision boundary between the two classes. Blue and orange points denote the two classes; the black curve is the equal-distortion boundary.

Figure 4 shows the corresponding test classification plot generated by the submitted notebook. Correctly classified samples and any misclassified samples are identified by the plot legend.

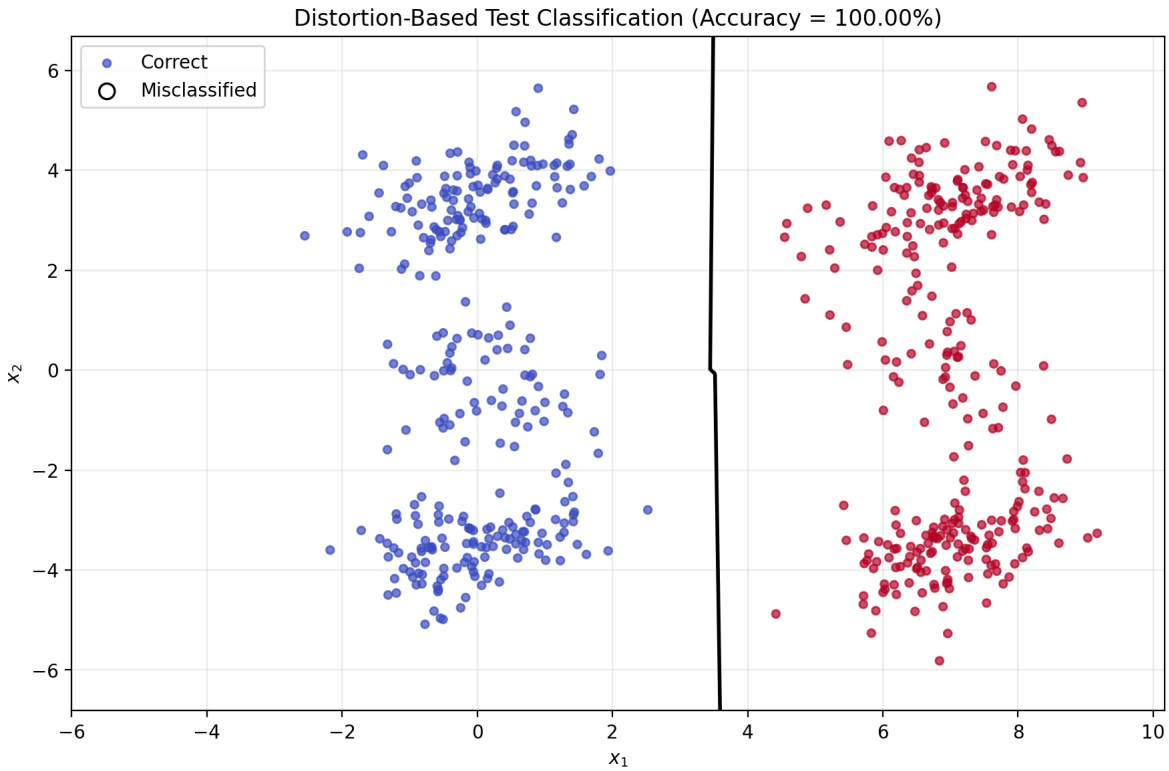


Figure 4: Notebook-generated distortion-based test classification and decision boundary.

4.3 Covariance comparison

The notebook reports 100.00% test accuracy for spherical, diagonal, and full covariance in this run.

Table 3: Test accuracy by covariance type

Covariance type	Test accuracy	Error rate
Spherical	100.00%	0.00%
Diagonal	100.00%	0.00%
Full	100.00%	0.00%

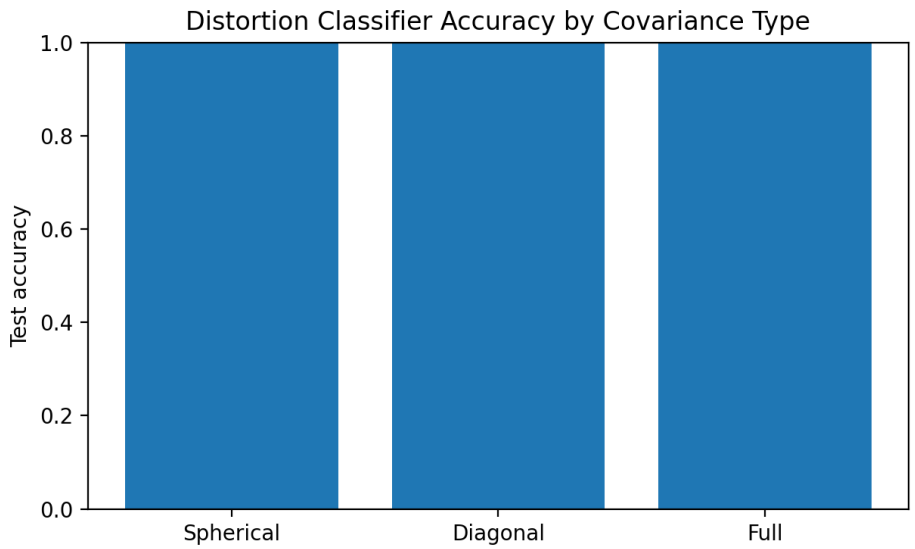


Figure 5: Distortion-classifier test accuracy by covariance type.

5 Answers to Assignment Questions

Question 1: Choose K, π_i, μ_i, C_i empirically.
Three Gaussian components are used for each class with weights $0.40, 0.20, 0.40$. Class 1 means are $(0, 3.5), (0, 0), (0, -3.5)$; Class 2 means are $(7, 3.5), (7, 0), (7, -3.5)$. The covariance matrices are listed in Section 2.

Question 2: Experiment with spherical, diagonal, and full covariance matrices.
All three covariance structures are generated and plotted in Figure 1.

Question 3: Plot points belonging to different clusters using different colours.
Figure 1 uses different colours for the three Gaussian components in each class.

Question 4: Generate data for two classes, each represented by a mixture density.
Two classes of 1,500 observations each are generated, with three Gaussian components per class and horizontal separation.

Question 5: Divide data 80:20, perform K-Means on 80%, and plot cluster boundaries.
Each class has 1,200 training and 300 test observations. K-Means uses $K = 2$ centroids per class; Figure 2 shows the within-class regions.

Question 6: Classify the remaining 20% using distortion.
Each test sample is assigned to the class with the smaller minimum squared centroid distance. The main run gives 100.00% accuracy and confusion matrix $\begin{bmatrix} 300 & 0 \\ 0 & 300 \end{bmatrix}$.

Question 7: Report observations.
Covariance controls cluster shape; K-Means learns class-specific prototypes, and distortion classification assigns each test point to the closest prototype. The classes are well separated along x_1, explaining the high accuracy in this synthetic experiment.

6 Observations

1. Box–Muller converts uniform random values into standard-normal samples.
2. Covariance eigendecomposition is used; Cholesky decomposition is not used.
3. Each class uses three Gaussian components with weights $0.40, 0.20, 0.40$.

4. Spherical, diagonal, and full covariance matrices produce different cluster geometries.
5. K-Means uses two centroids per class, distinct from the three Gaussian components used for generation.
6. The distortion classifier achieves 100.00% accuracy in the main full-covariance run; all three covariance experiments report 100.00% in this run.
7. Results depend on the selected parameters, generated samples, and centroid initialization.

7 Conclusion

Two two-dimensional Gaussian-mixture classes were generated and studied under spherical, diagonal, and full covariance assumptions. The data were split in an 80:20 ratio, K-Means was fitted separately to each class, and held-out observations were classified using minimum squared distortion. The main run correctly classified all 600 test observations. The experiment demonstrates how covariance affects cluster geometry and how class-specific K-Means centroids can be used as prototypes for classification.